



Precision, Low Power INSTRUMENTATION AMPLIFIERS

FEATURES

- LOW OFFSET VOLTAGE: 50 μ V max
- LOW DRIFT: 0.5 μ V/ $^{\circ}$ C max
- LOW INPUT BIAS CURRENT: 5nA max
- HIGH CMR: 120dB min
- INPUTS PROTECTED TO $\pm$ 40V
- WIDE SUPPLY RANGE: $\pm$ 2.25V to $\pm$ 18V
- LOW QUIESCENT CURRENT: 700 μ A
- 8-PIN PLASTIC DIP, SO-8

APPLICATIONS

- BRIDGE AMPLIFIER
- THERMOCOUPLE AMPLIFIER
- RTD SENSOR AMPLIFIER
- MEDICAL INSTRUMENTATION
- DATA ACQUISITION

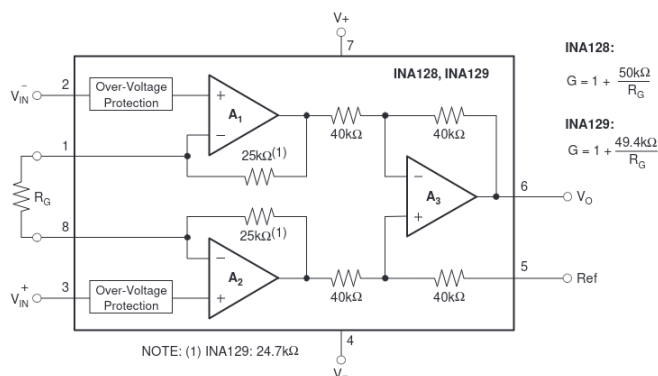
DESCRIPTION

The INA128 and INA129 are low power, general purpose instrumentation amplifiers offering excellent accuracy. The versatile 3-op amp design and small size make them ideal for a wide range of applications. Current-feedback input circuitry provides wide bandwidth even at high gain (200kHz at $G = 100$).

A single external resistor sets any gain from 1 to 10,000. The INA128 provides an industry-standard gain equation; the INA129 gain equation is compatible with the AD620.

The INA128/INA129 is laser trimmed for very low offset voltage (50 μ V), drift (0.5 μ V/ $^{\circ}$ C) and high common-mode rejection (120dB at $G \geq 100$). It operates with power supplies as low as $\pm$ 2.25V, and quiescent current is only 700 μ A—ideal for battery-operated systems. Internal input protection can withstand up to $\pm$ 40V without damage.

The INA128/INA129 is available in 8-pin plastic DIP and SO-8 surface-mount packages, specified for the -40° C to $+85^{\circ}$ C temperature range. The INA128 is also available in a dual configuration, the INA2128.



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